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DIGITAL PRODUCT PASSPORT · MULTI-BODY STEWARDSHIP

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ISO/IEC JTC 5 · DIGITAL PRODUCT PASSPORT

Semantic registry requirements for the Digital Product Passport

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UniDPP — convened under CalConnect, multi-body stewardship

01 · SCOPE

What "filling §4.3" means

This contribution specifies the requirements for the semantic registry of the Digital Product Passport ecosystem: the governance model, the identifier discipline, the content layout, the resolution protocol, the exchange serialization, and the conformance regime. It is offered as input to JTC 5 workstreams B and C, and as the technical basis of a new-work-item proposal (NWIP-2) co-sponsored by ISO/TC 154, UN/CEFACT and the FERIN operators.

The requirement source is the vacancy itself. EN 18223 §4.3 makes a semantic repository *required*, *governed* and *"regulation-based"*, and then leaves the operator, the protocol, resolution, caching, offline behaviour and failure semantics unspecified; the clause's examples are placeholder URLs (`dictionary1.eu`). The vacancy is now confirmed from the payload side by machine verification: a run of every example payload printed in EN 18223 (22 fixtures; paper 4, section 05) finds every Annex A `dictionaryReference` aimed at the placeholder host — repository, operator and resolution protocol unspecified (AUDIT C1) — and Annex A Example 3 printing `"https:/dictionary1/maximumPressure"`, a reference that omits the URI authority separator and cannot resolve even in principle (finding N2). Every `dictionaryReference` in the EN series resolves into this vacancy. The surrounding standard, meanwhile, specifies serialization detail in abundance — per-class JSON construction rules in §5.2.5–5.2.9, identifier casing in §5.2.2, XSD type exclusions in §4.1.2.9 — while the meanings those serializations carry have no registered home. The class taxonomy of EN 18223 (Single/MultiValued, MultiLanguage, RelatedResource, Collection) classifies *shapes of records*, not meanings of data.

> **Design position.** The registry is not an accessory to the passport; it is the component that makes multi-jurisdiction operation possible. Jurisdictions that regulate differently must register **distinct data elements harmonized to a common concept** rather than overwrite one definition — divergence becomes registered, auditable content instead of textual contradiction.

02 · GOVERNANCE

Federated registers per ISO 19135:2025

The governance model is the procedures standard already written for exactly this problem: ISO 19135:2025, *Geographic information — Procedures for item registration*, operated as FERIN (the federated register model the EPSG and OGC registers have run at global scale for decades). The semantic registry shall be a **principal register with federated subregisters**:

19135 role	Responsibility in the DPP registry
Register owner	Sets scope and policy of a (sub)register; for jurisdictional subregisters, typically the national standards body or regulator-nominated authority.
Register manager	Day-to-day operation, item admission workflow, publication of the register in machine-readable form.
Submitting organization	Proposes items: regulators and their NSBs for jurisdiction profiles; sector bodies for overlays; standards committees for base data elements.
Control body	Decides proposals; owns the harmonization rulings that map divergent regional items to common concepts.

Item lifecycle follows 19135: **proposal → public review → valid**, with **superseded** and **retired** as the only exits — items are never edited in place; a corrected definition is a new version whose predecessor remains resolvable, because every issued passport pins the version it was validated against. Profiles constrain or add to registered definitions; they never redefine them.

The subregister layout and the UNTDED seed

The principal register federates at least these subregisters:

- **Data elements** — seeded from **UNTDED**, the United Nations Trade Data Element Directory (live at untded.org; ISO/TC 154 document N1727), with national registration authorities as federated subregisters, per the historic UNTDED practice of national regional subsets;
- **Jurisdiction and sector profiles** — the registered profile objects of the framework's lens model (paper 1), including their trigger predicates and effective dates; this subregister also carries **carrier and presentation profiles** with their translation rules: GB/T 33993 is a national wrapper over GS1 syntax, so cross-carrier interoperability needs registered translation rules (QR code structure, mandatory keys, URL patterns) between GB/T 33993, GS1 Digital Link and the EN 18220 carrier bindings;
- **Cryptographic suites** — per-jurisdiction algorithm bindings (SM-family as ISO/IEC 14888-3/10118-3/18033-3 mechanisms; FIPS suites; post-quantum families) with migration phases;
- **Trust anchors and the master list** — the registry content of the cross-recognition trust architecture (paper 5);
- **Units and quantities** — the ISO 80000 → UnitsDB/UnitsML chain, closing the unit-binding gap that runs through the whole EN series.

Identifier discipline: items are identified per **ISO/IEC 15459** (with IRDI retained as a legacy alias), and every reference is a `(register, item, version)` triple with one canonical resolvable URI form. Cross-register mappings (e.g., an EN data element to its UNTDED and UNTP counterparts) are themselves registered items under the 19135 harmonization mechanism — mapping is content, not prose. The same mechanism carries two further mapping classes observed to be load-bearing in deployed systems. **Standard-equivalence mappings**: certificates cite national standards (GB 4943.1-2022 ≈ IEC 62368-1), and a foreign verifier needs a registered equivalence claim with an attestor — conformity-evidence translation is a profile/transform problem, not prose. And **carrier-profile translation rules** as registered transforms, so that a passport resolvable through one national code format verifies against another's expectations without re-issuance.

Finally, the registry carries **dormant identifiers**. Inputs are recorded at the finest identifier granularity available — (scheme, lot or serial, count) — even where no passport yet exists behind them: when a future regime issues passports for that object class (cells, modules, materials), it **adopts the recorded dormant identifiers** rather than minting new ones and orphaning the installed base. Dormant → live adoption connects the entire installed base retrospectively through build records; retrospective issuance proceeds issue-by-records or issue-on-first-touch, with retroactive applicability already carried by dated bindings (invariants I1 and I3 of paper 1).

04 · RESOLUTION PROTOCOL

The dictionaryReference contract

The vacancy of §4.3 is filled by making `dictionaryReference` a protocol, not a URL convention. Resolution of a canonical item URI returns a signed bundle comprising:

1. the **definition** (human-readable, multilingual where the source register provides it) and its structured machine form;
2. the exact **version**, with its predecessor/successor chain;
3. the register **status** (valid / superseded / retired) and effective dates — enabling "was this definition in force when the passport was validated" queries;
4. the **trust chain** binding the item to its register's signing authority.

Representation is by content negotiation among **CDDAL (plain text), JSON-LD and XML**. Versioned definitions are immutable and therefore perfectly cacheable; mirrors and national hosts replicate them freely, carrying log-anchored inclusion proofs so a verifier never depends on the origin server's availability. Offline and failure semantics follow the framework's degradation ladder: cached definitions with as-of stamps, evidentiary mode when anchors cannot be reached. Profiles pin registry versions at verification time — deterministic replay of a validation requires the same definitions that were in force at the audited moment.

05 · DICTIONARY TECHNOLOGY

CDDAL canonicalization and machine validation

Dictionaries will not be monolithic: multiple hosts — including national ones — hold CDD/IEC 61360-62656-class content. Exchange between hosts uses **CDDAL** as the plain-text serialization of that content, with a **normative canonicalization rule** so that a definition's hash is stable across hosts: two mirrors of item version `(R, D, v)` produce byte-identical canonical forms. Cross-dictionary comparison, conformance tooling and audit all reduce to hashing. OpenCDD (operated by ELF) provides the open reference implementation of the stack; EXPRESS-encoded content is validated with expressir; whole semantic models are exchanged as registered, checksummed artifacts.

> **Why registration must be machine-checked.** The converted corpus shows that hand-maintained, machine-resolvable identifiers do not survive editorial processes unassisted: published consortium specifications contain semanticId fields with typographical defects, broken path segments, and swapped identifiers, and one specification's mapping table cites "EN 1822:2026" — a missing digit in a normative citation. None of these would survive an automated lint at registration. Register admission shall therefore include syntactic validation, reference resolution, and canonicalization round-trip as mandatory gates — the same discipline EN 18223 asks of others applied to the registry itself.

Consolidated statements

The following statements are proposed as the normative core of NWIP-2. "Shall" expresses a requirement on any conforming registry deployment.

ID	Requirement
R1	The registry **shall** operate as a principal register with federated subregisters under the governance roles and item lifecycle of ISO 19135:2025 (FERIN).
R2	Every item **shall** carry an identifier per ISO/IEC 15459 and be referenceable as a (register, item, version) triple with one canonical resolvable URI; IRDI is supported as a legacy alias.
R3	Item versions **shall** be immutable; correction occurs by supersession, and superseded versions remain resolvable with status and effective dates.
R4	The data-element subregister **shall** be seeded from UNTDED, with national registration authorities as federated subregisters.
R5	`dictionaryReference` resolution **shall** return definition, exact version, register status and trust chain, by content negotiation over CDDAL, JSON-LD and XML.
R6	Profiles and passports **shall** pin registry item versions at validation/verification time, enabling deterministic as-of replay.
R7	Divergent regional regulations **shall** be registered as distinct items harmonized to a common concept by registered cross-register mappings; redefinition of another jurisdiction's item is not permitted.
R8	Subregisters **shall** exist for data elements, jurisdiction/sector profiles, cryptographic suites, trust anchors/master-list entries, and units and quantities (ISO 80000 → UnitsDB/UnitsML).
R9	Host-to-host exchange **shall** use CDDAL with a normative canonicalization rule producing host-stable definition hashes.
R10	Mirrors **shall** serve cached definitions with log-anchored inclusion proofs; verification **shall not** depend on the availability of the origin host.
R11	Register admission **shall** enforce machine validation: syntax, reference resolution, and canonicalization round-trip.
R12	Conformance **shall** be defined by conformance classes (register operator, subregister operator, mirror, client) with a published executable test suite.

Two boundaries are equally normative. First, the registry registers *meanings, their versions and their mappings* — it does not host passports, keys or events; those live in the resolution, trust and event layers with which it is federated. Second, the semantic registry and the trust registry (paper 5) are siblings under the same 19135 registration discipline: one governs what data means, the other governs who is trusted to say it. Neither subordinates to the other.

07 · CONFORMANCE AND PILOT

Proving the requirements executable

The requirements are accompanied by an open conformance and demonstration stack: the FERIN register operations (Enosema) as operator reference; OpenCDD as dictionary technology (ELF); UNTDED as seed content (UN/CEFACT / ISO/TC 154 N1727); expressir validation for EXPRESS content; and registry clients in the UniDPP reference implementation (paper 4) exercising R1–R12 against a public test register. The pilot proposed to JTC 5 is deliberately small: register the EN 18223 data-element set as versioned items with UNTDED mappings, add one divergent jurisdiction item (a substance-restriction list entry) with its harmonization mapping, and demonstrate as-of resolution across a supersession — the minimum evidence that the vacancy of §4.3 is closed, not merely described.

08 · REFERENCES

1. EN 18223, Digital Product Passport — semantic model, §4.1, §4.3, §5.2 (CEN-CLC/JTC 24).
2. ISO 19135:2025, Procedures for item registration (FERIN federated register model).
3. ISO/IEC 15459 series, unique identifiers.
4. UNTDED, United Nations Trade Data Element Directory; ISO/TC 154 N1727.
5. IEC 61360 / ISO/IEC 62656, dictionary content model; CDDAL serialization; OpenCDD.
6. ISO 80000 quantities and units; UnitsDB; UnitsML.
7. ISO 10303-11 (EXPRESS) and expressir.
8. Companion papers 1, 4 and 5 of this series.

Drafting: ISO/TC 154, Enosema, UN/CEFACT, with ELF — in the UniDPP framework (multi-body stewardship; Apache-2.0/MIT per repository).

Paper 2 of 6 · semantic-registry-requirements.html · 2026-09-07